MS1Vtx_l2ecal_raw

overview · all regions and selections · rank 47 / 169

rank 47 / 169 by endcap / ht40 / D_equal

region	selection	D_equal	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg
endcap	all	0.343	0.349	0.352	0.336	0.344	0.344	0.349	0.363	34,970
endcap	ht40	0.431	0.437	0.441	0.418	0.432	0.431	0.437	0.450	7,390
endcap	bkg split	0.115	below 27,580 / above 7,390							
barrel	all	0.267	0.270	0.321	0.218	0.258	0.277	0.290	0.257	1,089
barrel	ht40	0.422	0.428	0.475	0.366	0.412	0.433	0.448	0.414	220
barrel	bkg split	0.206	below 869 / above 220							
all	all	0.359	0.366	0.370	0.334	0.359	0.359	0.374	0.375	36,059
all	ht40	0.449	0.456	0.459	0.424	0.449	0.448	0.463	0.466	7,610
all	bkg split	0.117	below 28,449 / above 7,610							

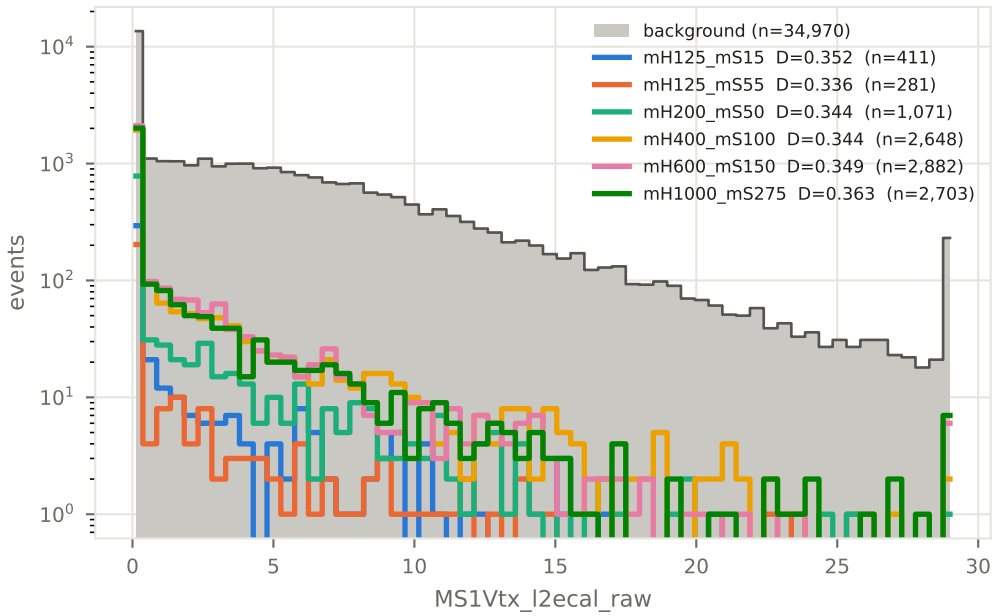
selections: all = every event; ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2

MS1Vtx_l2ecal_raw

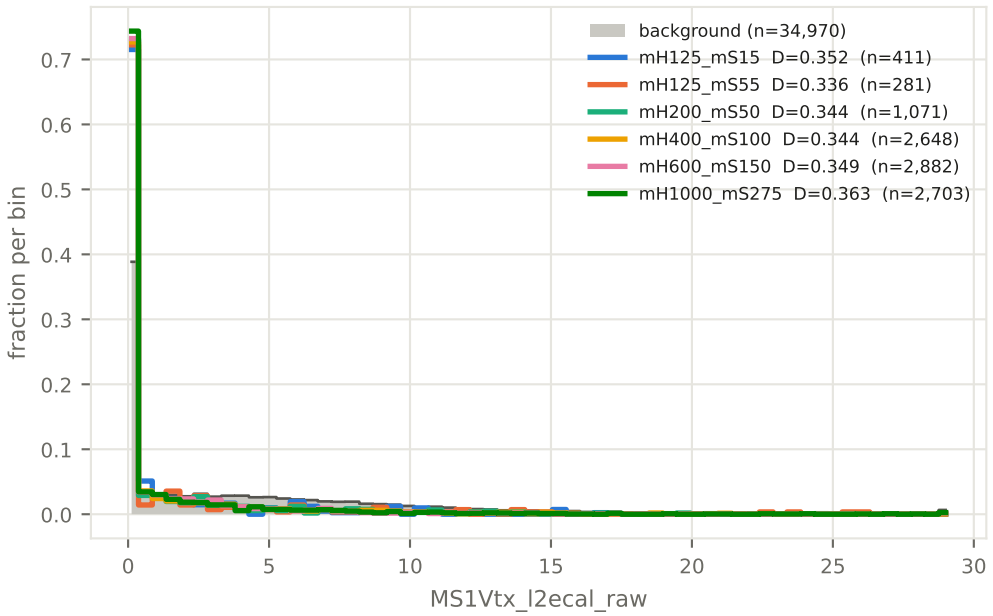
region: ENDCAP · selection: all (every event) · rank 47 / 169

D_equal 0.343 D_pooled 0.349 min 0.336 (mH125_mS55) max 0.363 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

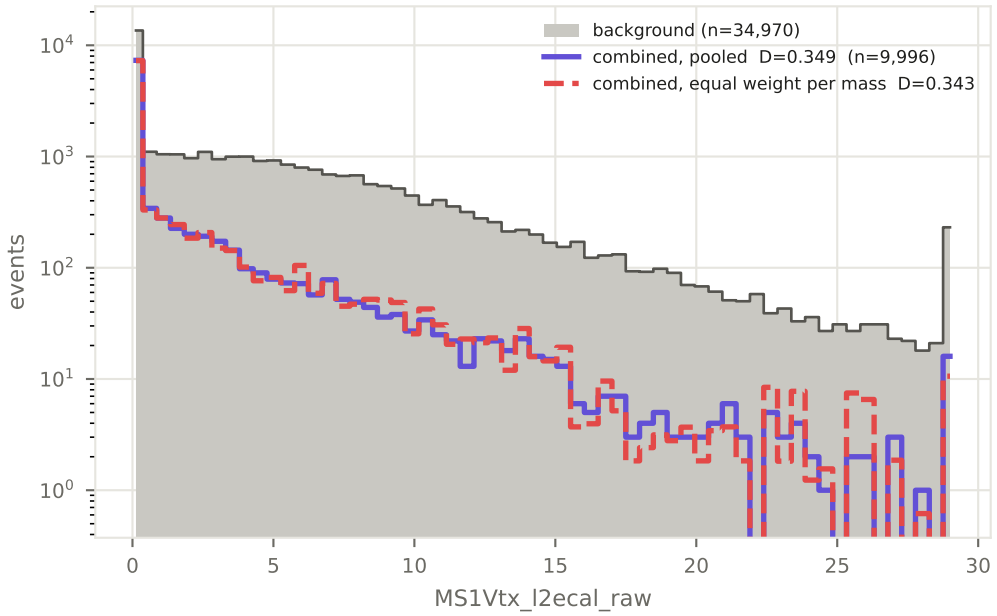
each mass point vs background · raw



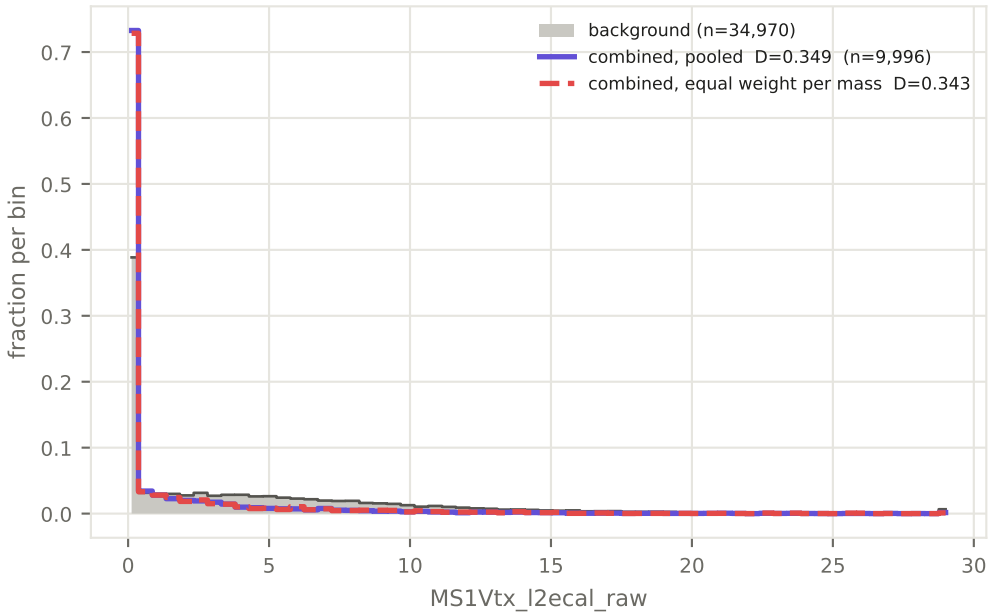
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



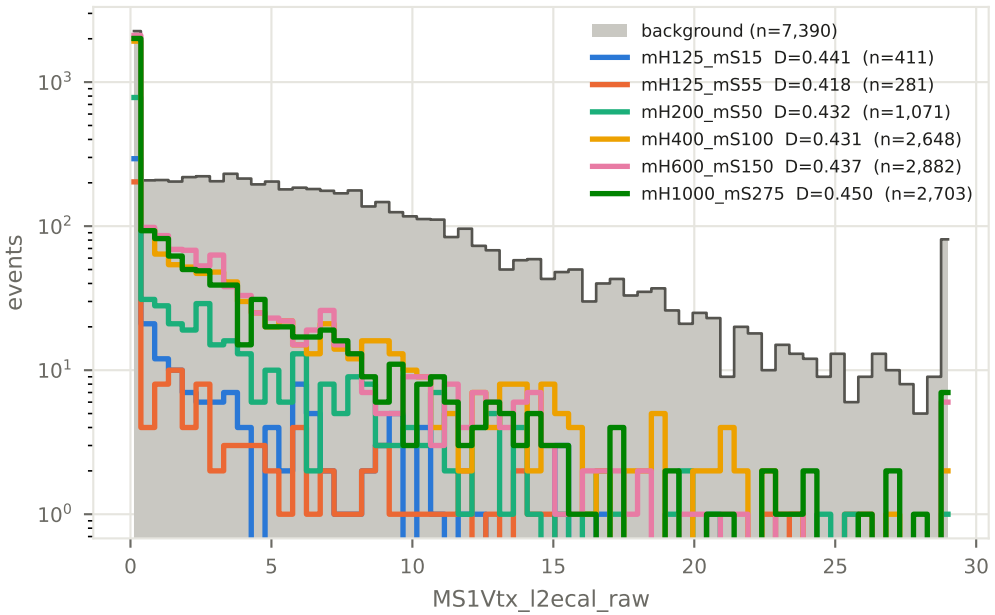
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_l2ecal_raw

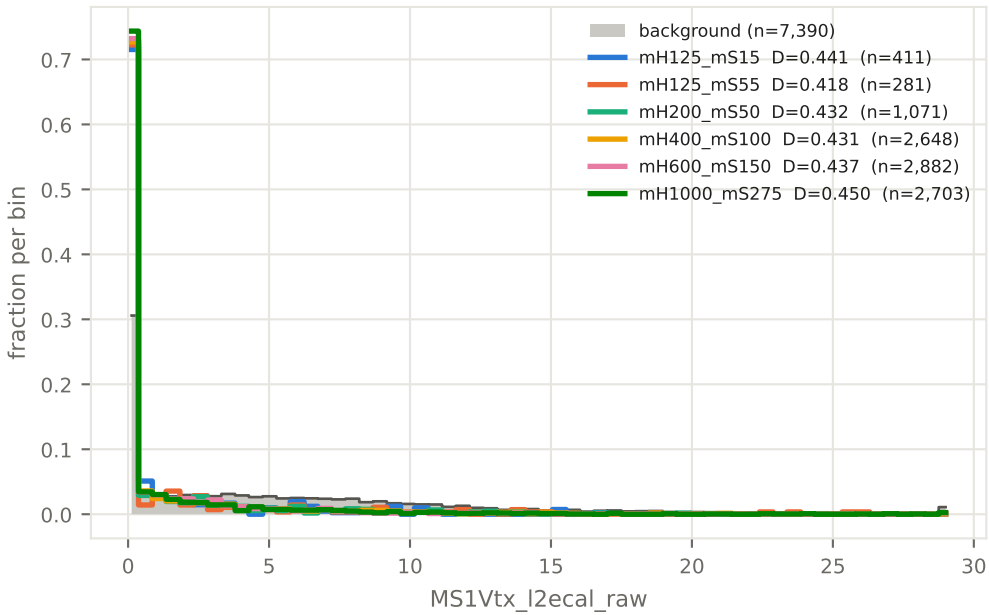
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 47 / 169

D_equal 0.431 D_pooled 0.437 min 0.418 (mH125_mS55) max 0.450 (mH1000_mS275) bkg n = 7,390 sig n = 9,996

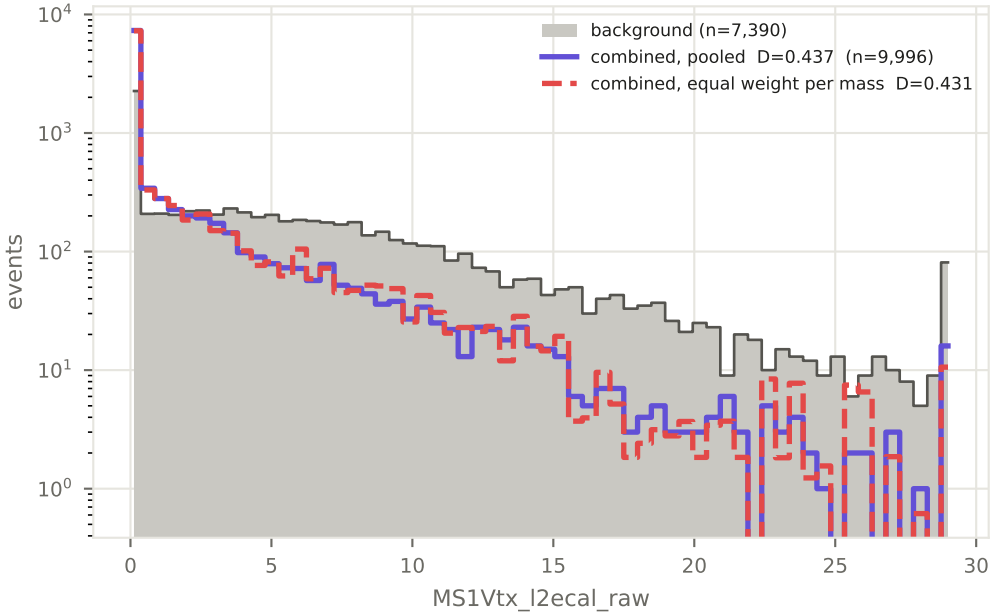
each mass point vs background · raw



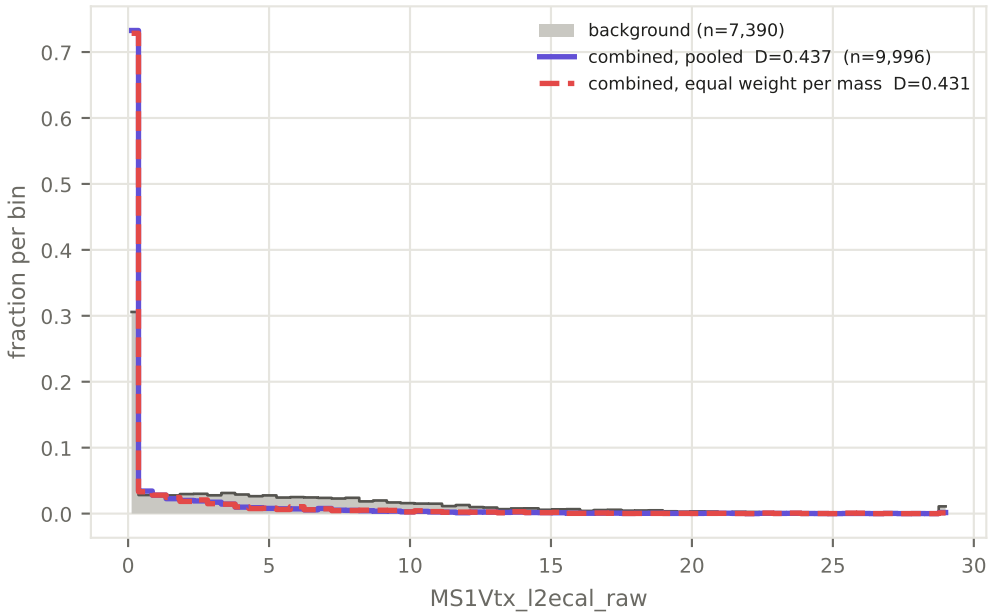
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

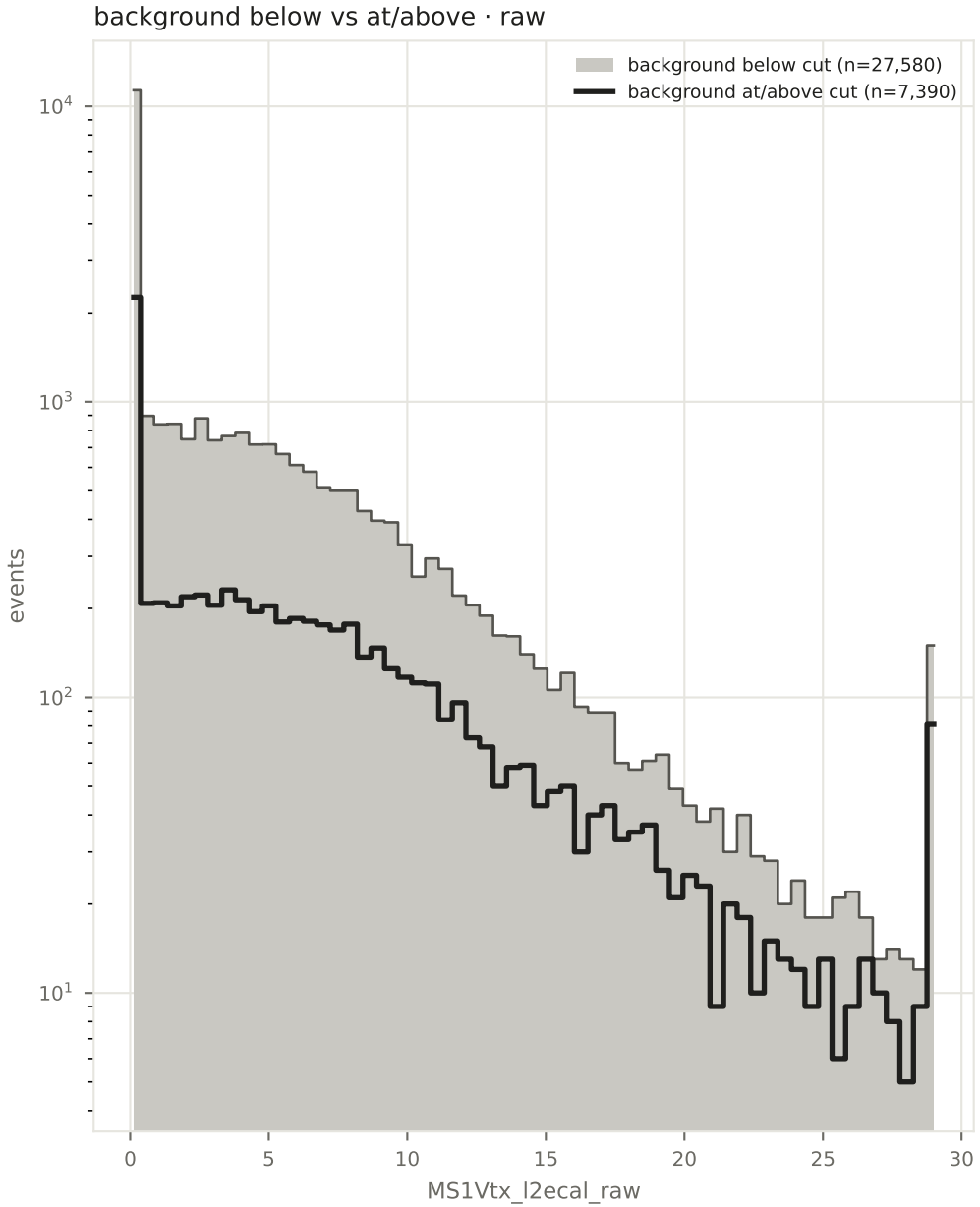


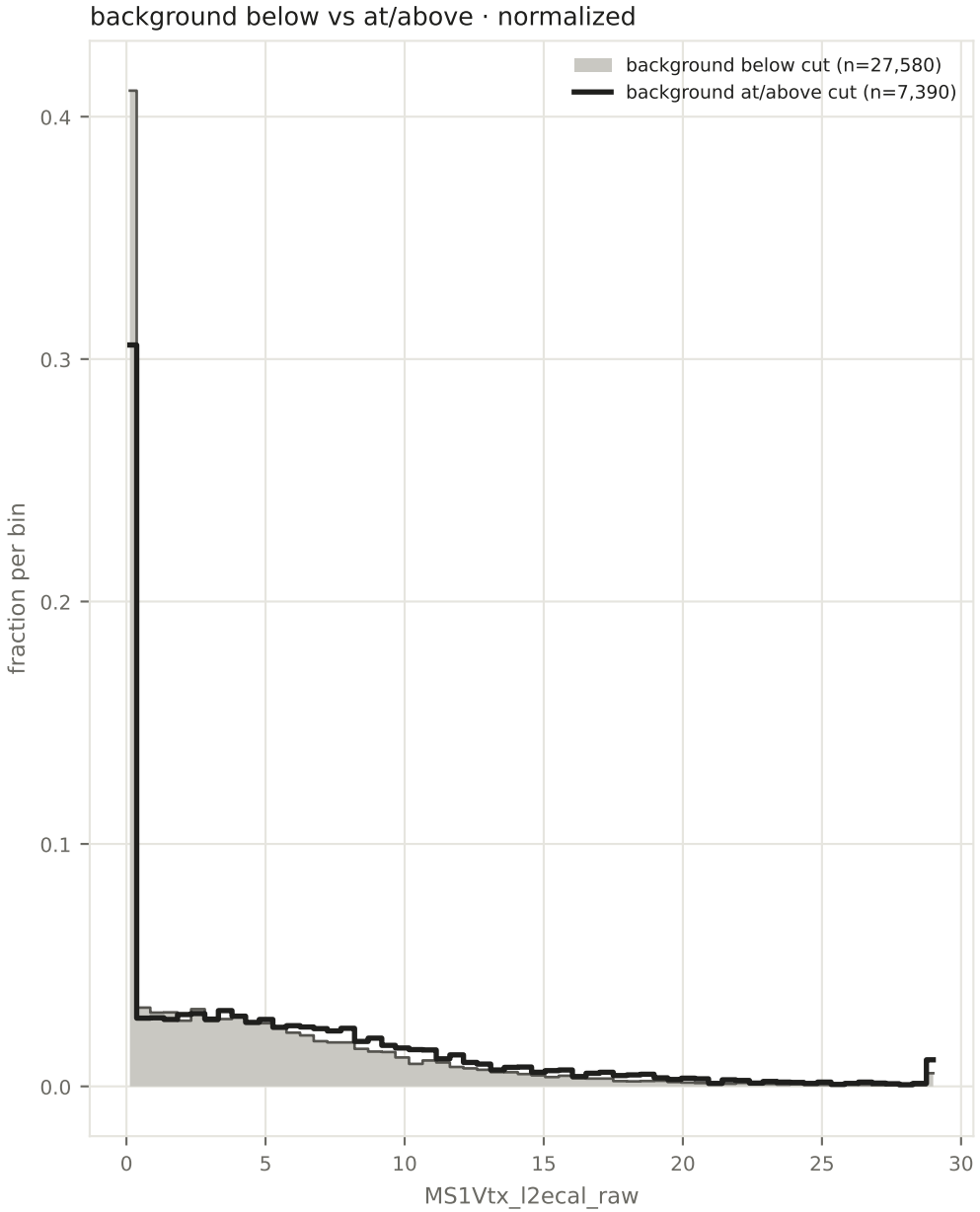
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_l2ecal_raw

region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 47 / 169

D(below vs above) 0.115 n below = 27,580 n above = 7,390 large D = background shape changes across the cut



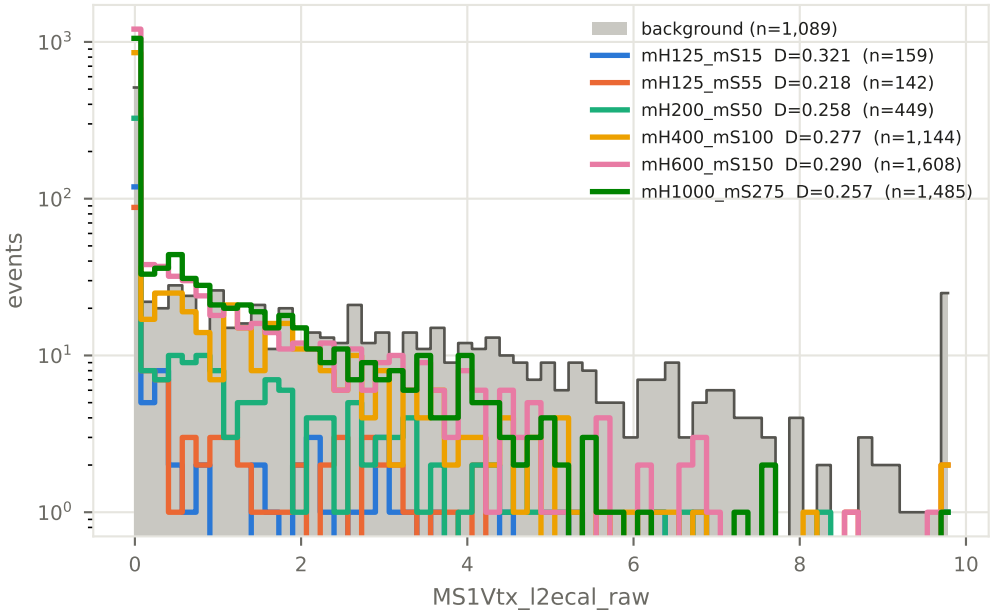


MS1Vtx_l2ecal_raw

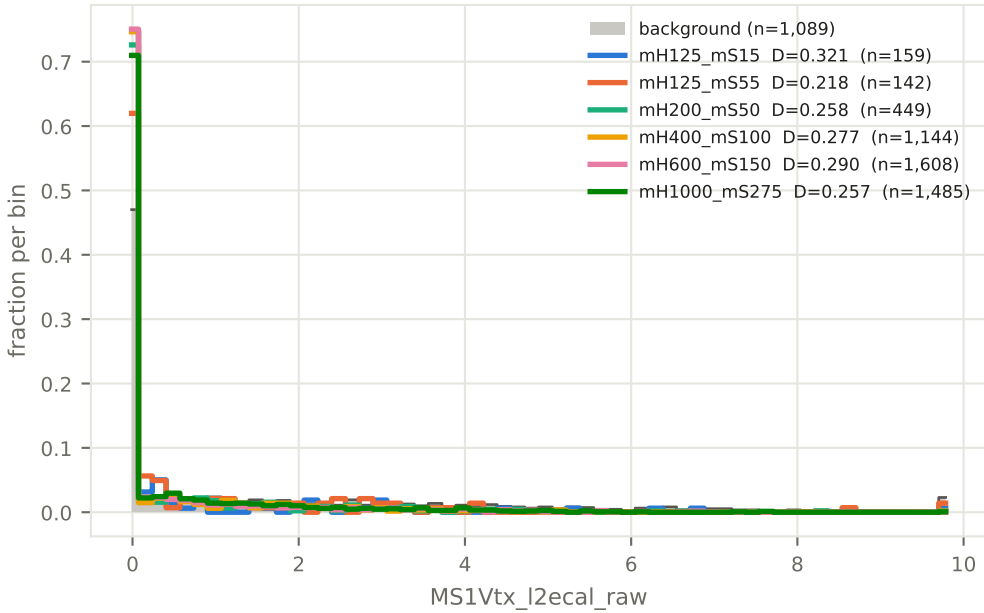
region: BARREL · selection: all (every event) · rank 47 / 169

D_equal 0.267 D_pooled 0.270 min 0.218 (mH125_mS55) max 0.321 (mH125_mS15) bkg n = 1,089 sig n = 4,987

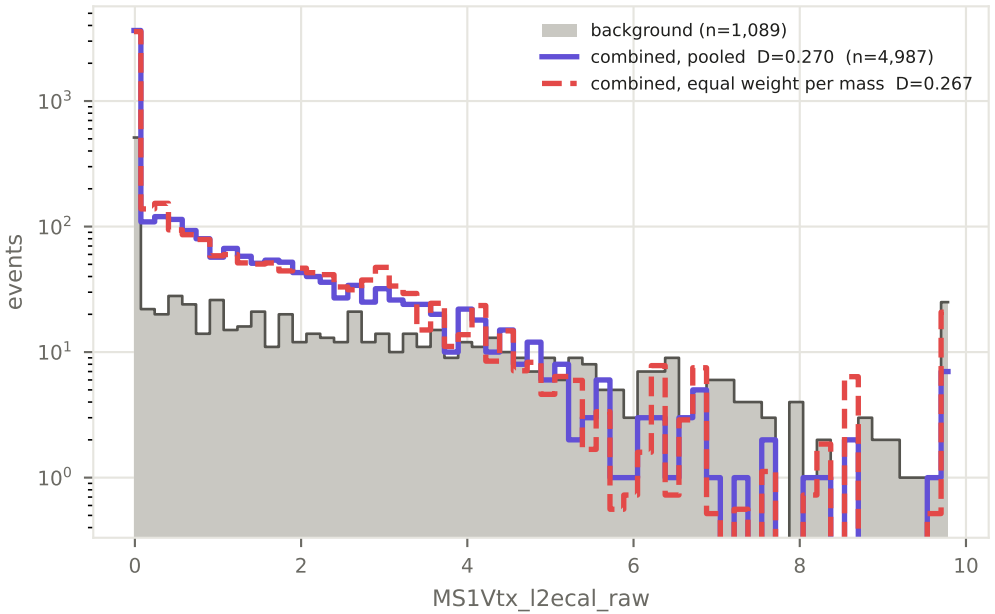
each mass point vs background · raw



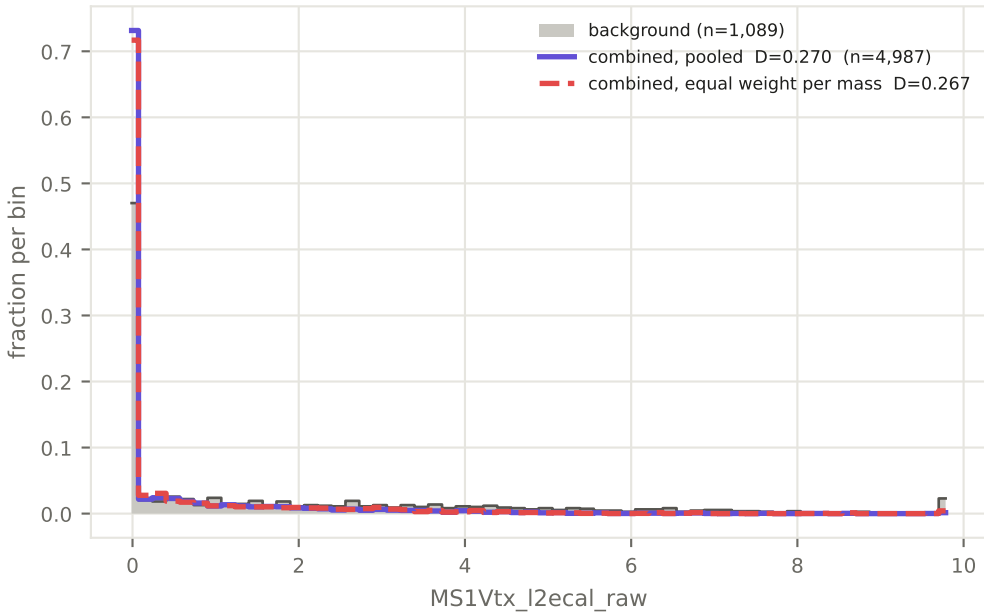
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



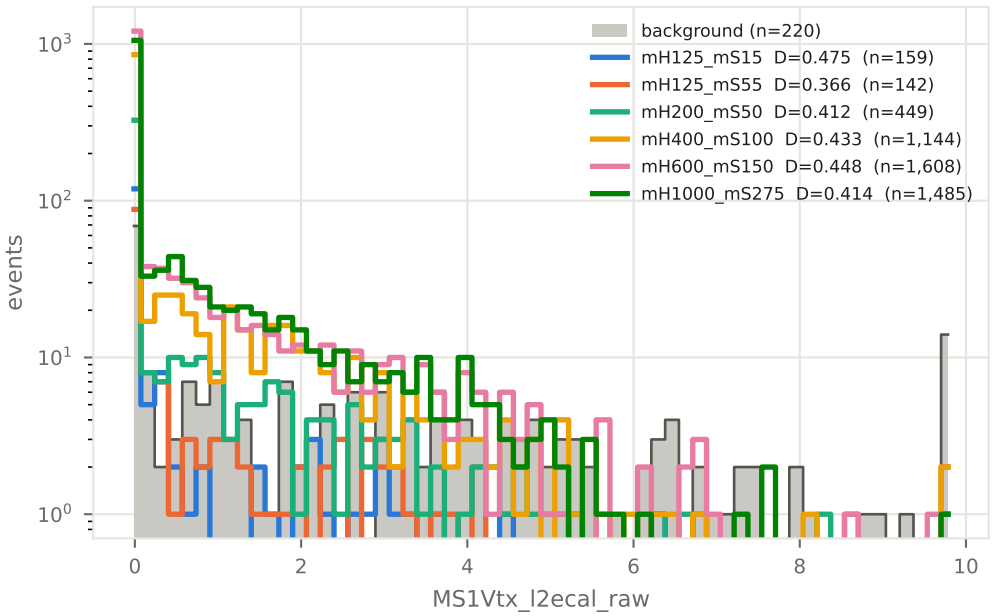
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_l2ecal_raw

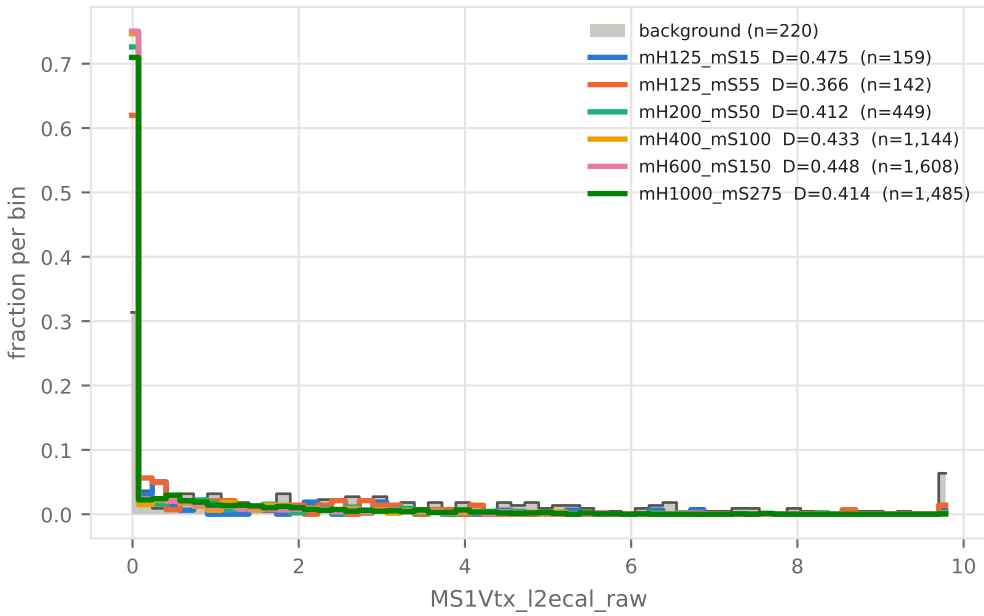
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 47 / 169

D_equal 0.422 D_pooled 0.428 min 0.366 (mH125_mS55) max 0.475 (mH125_mS15) bkg n = 220 sig n = 4,987

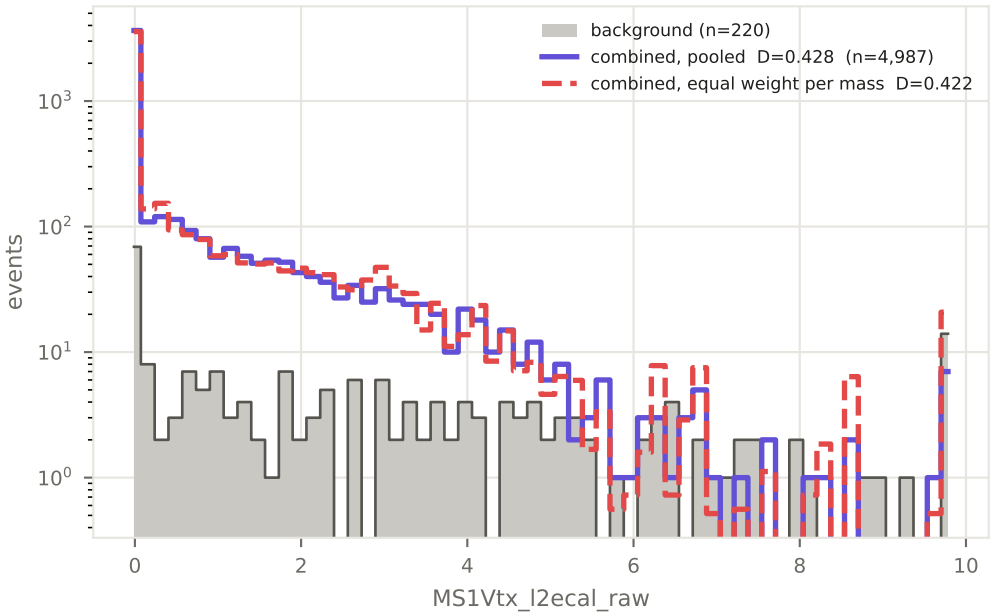
each mass point vs background · raw



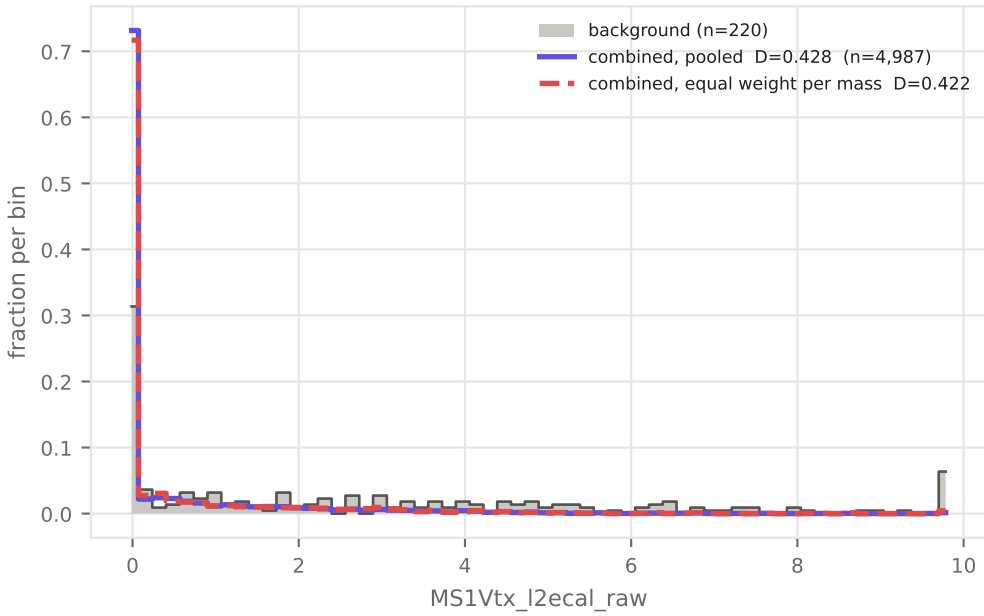
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

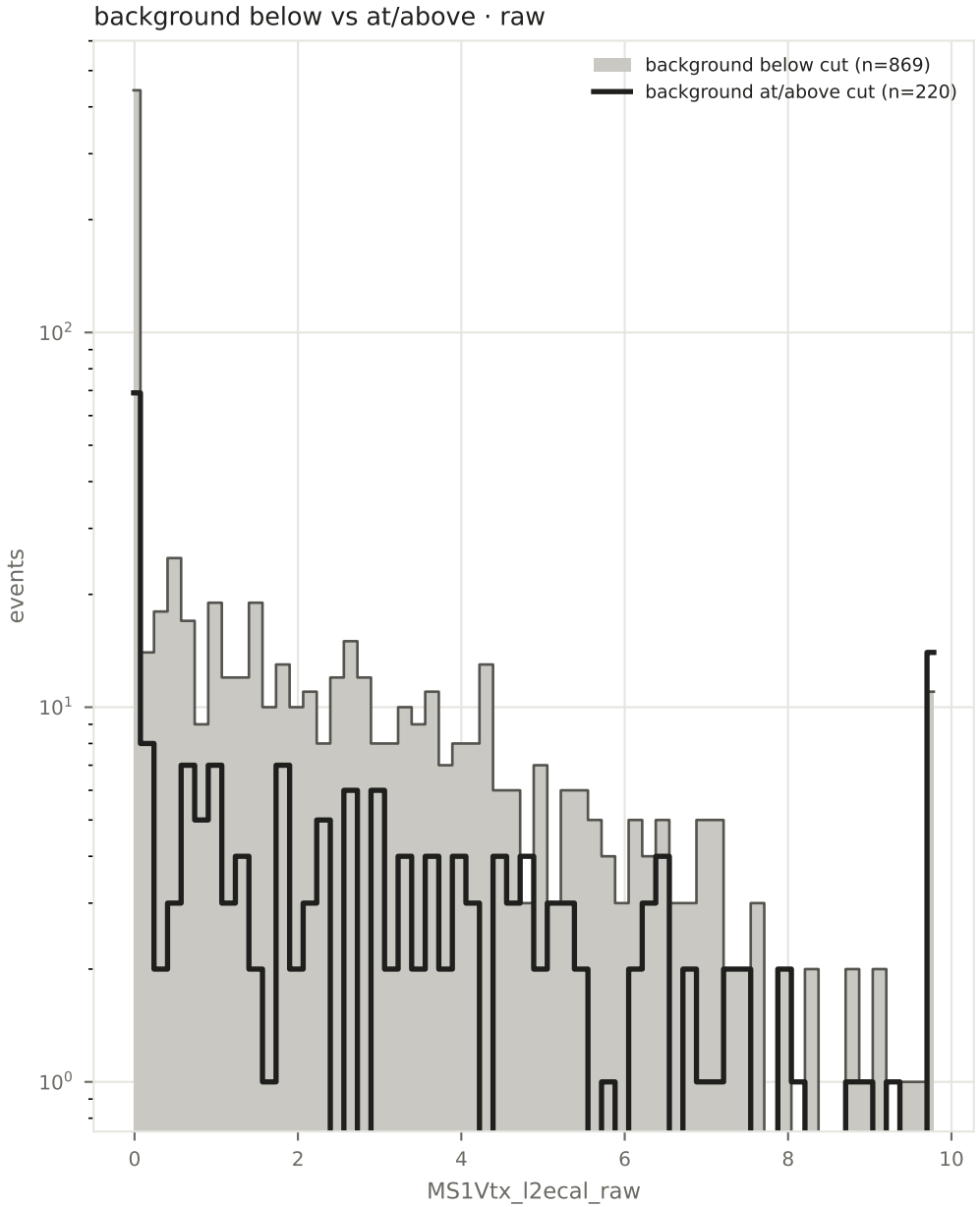


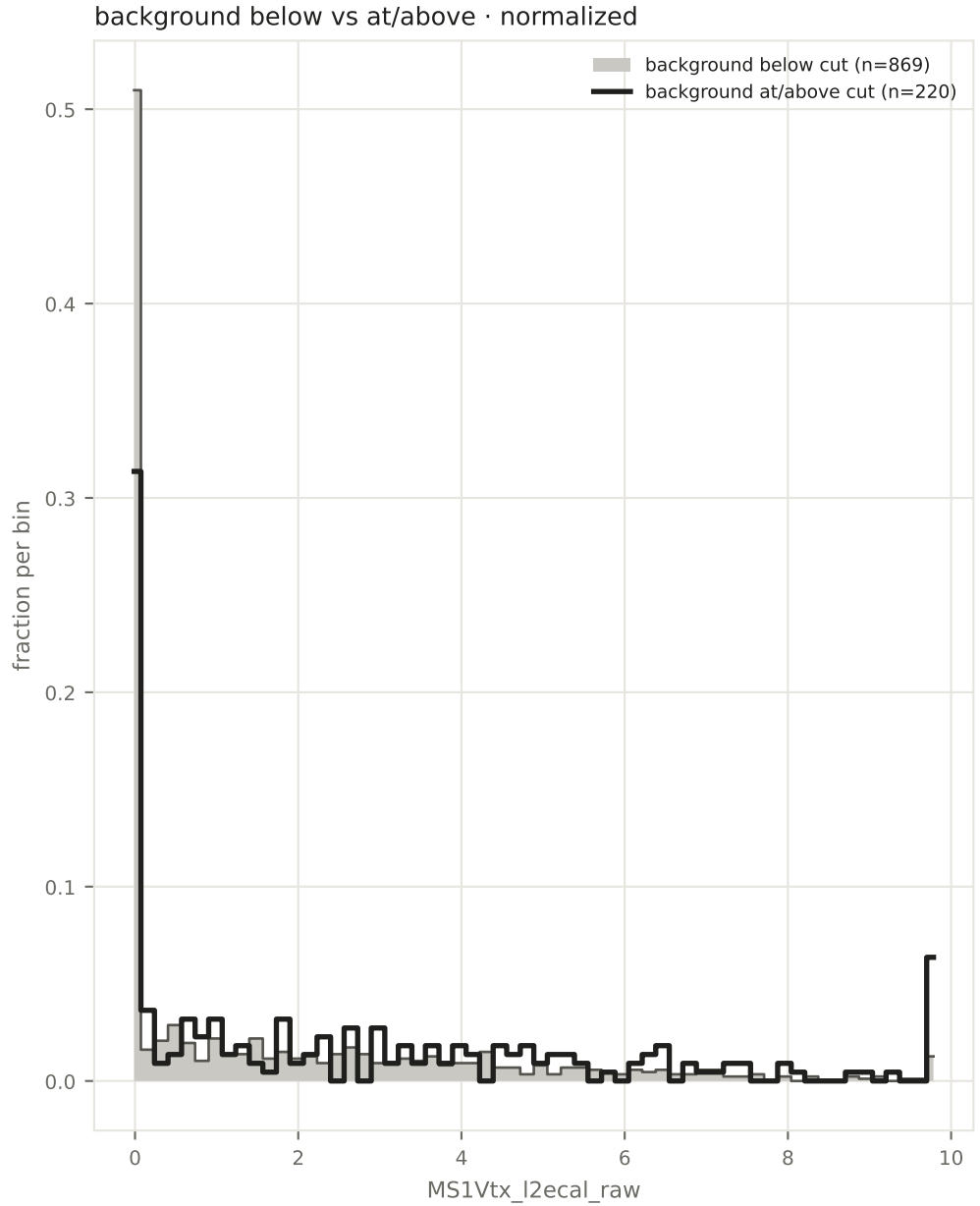
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_l2ecal_raw

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 47 / 169

D(below vs above) 0.206 n below = 869 n above = 220 large D = background shape changes across the cut



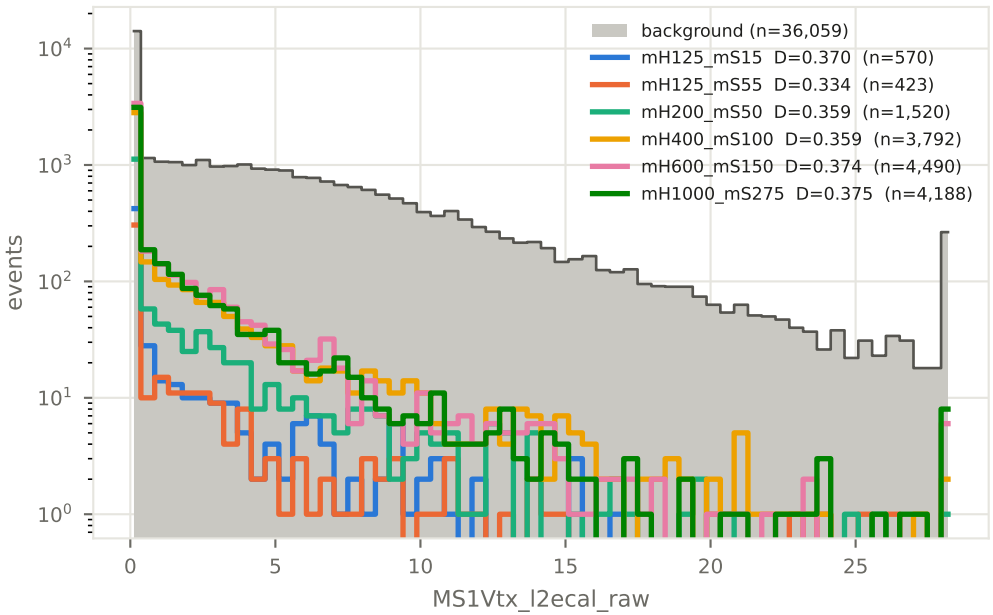


MS1Vtx_l2ecal_raw

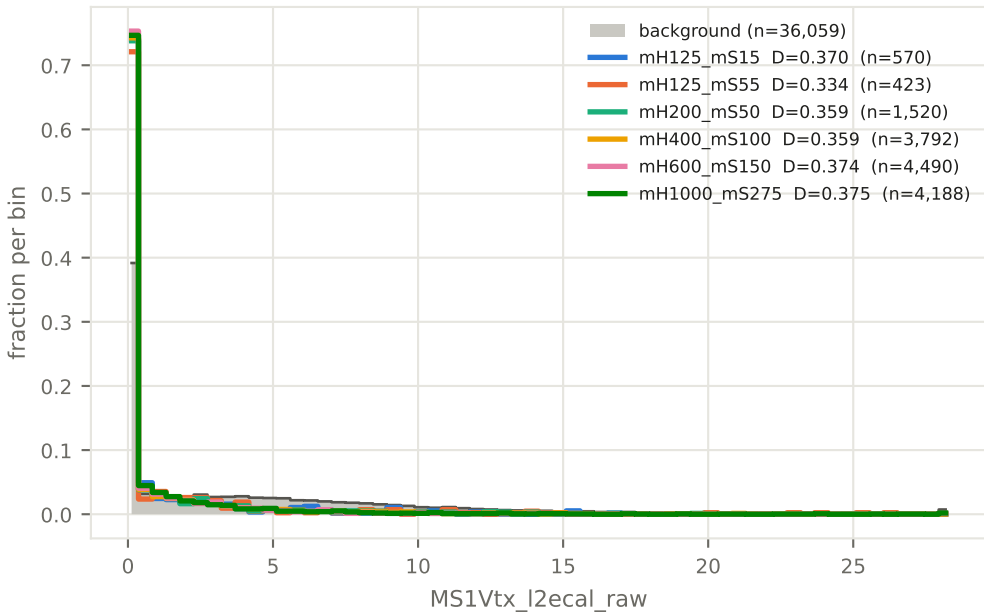
region: ALL · selection: all (every event) · rank 47 / 169

D_equal 0.359 D_pooled 0.366 min 0.334 (mH125_mS55) max 0.375 (mH1000_mS275) bkg n = 36,059 sig n = 14,983

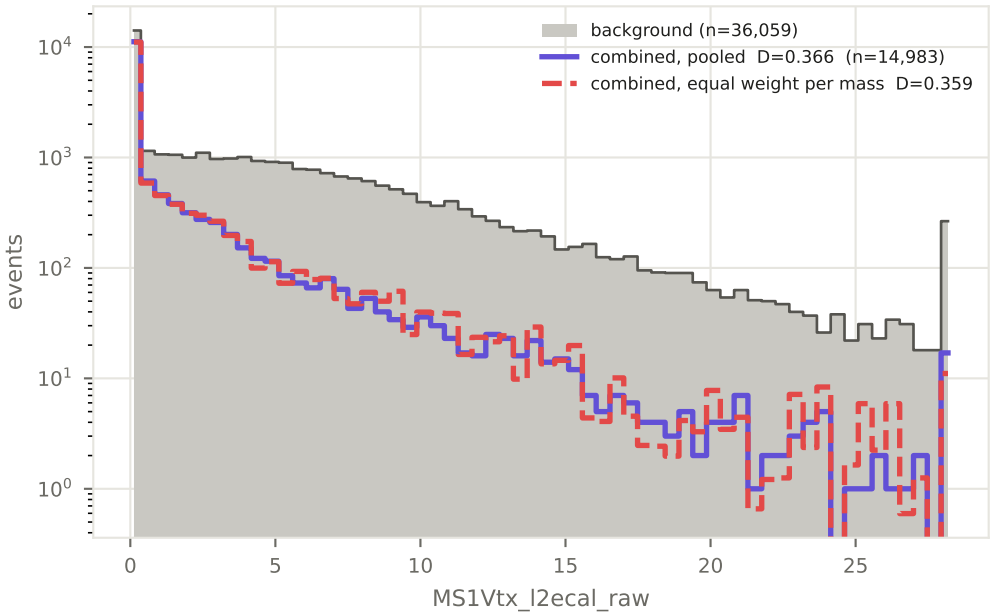
each mass point vs background · raw



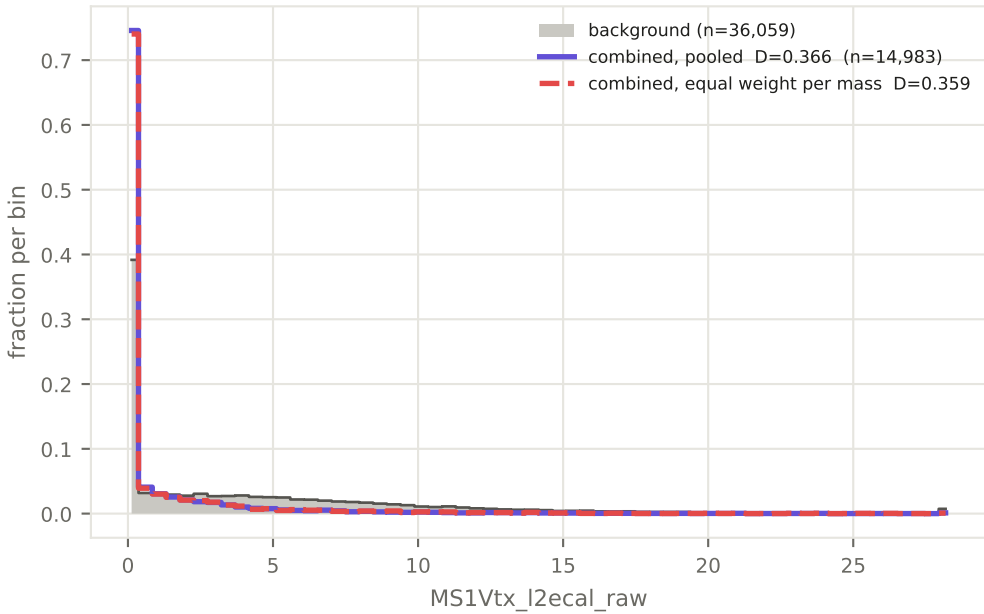
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



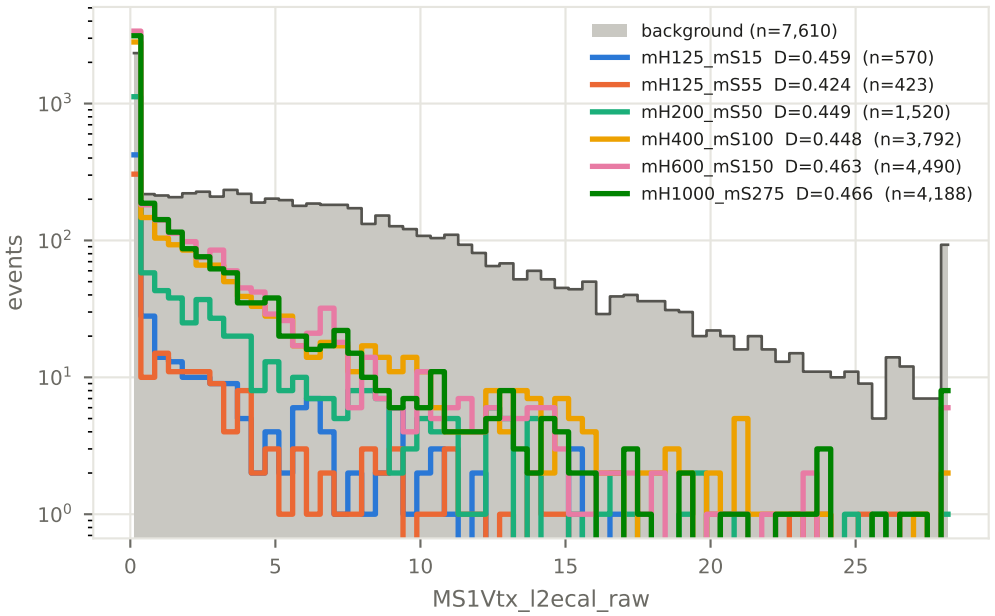
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_l2ecal_raw

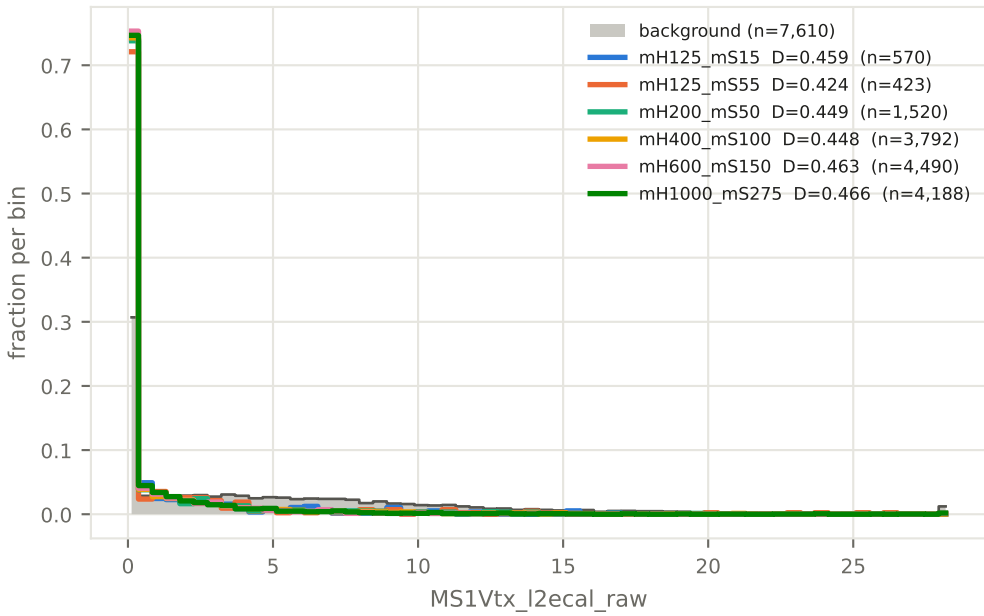
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 47 / 169

D_equal 0.449 D_pooled 0.456 min 0.424 (mH125_mS55) max 0.466 (mH1000_mS275) bkg n = 7,610 sig n = 14,983

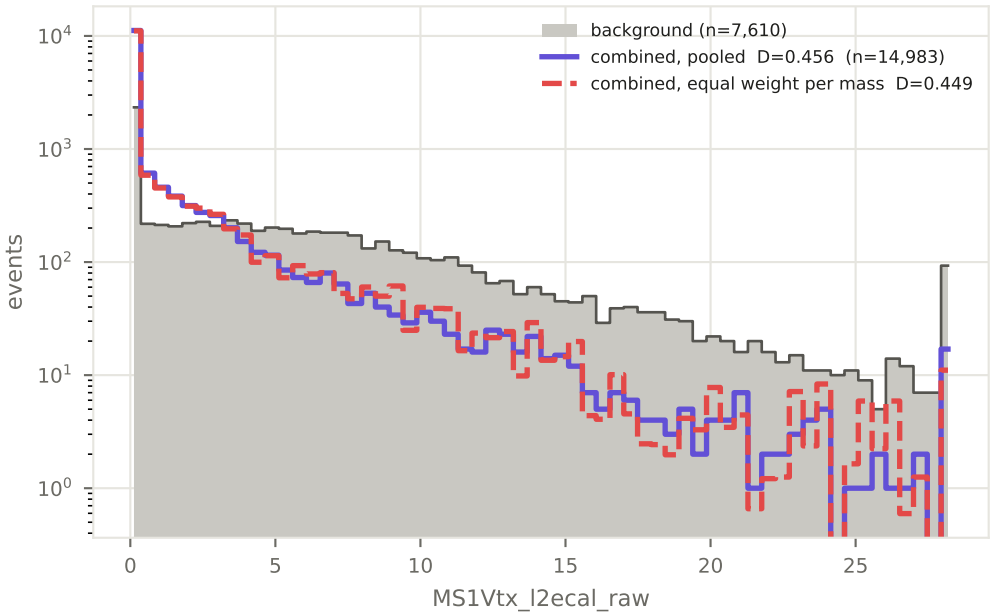
each mass point vs background · raw



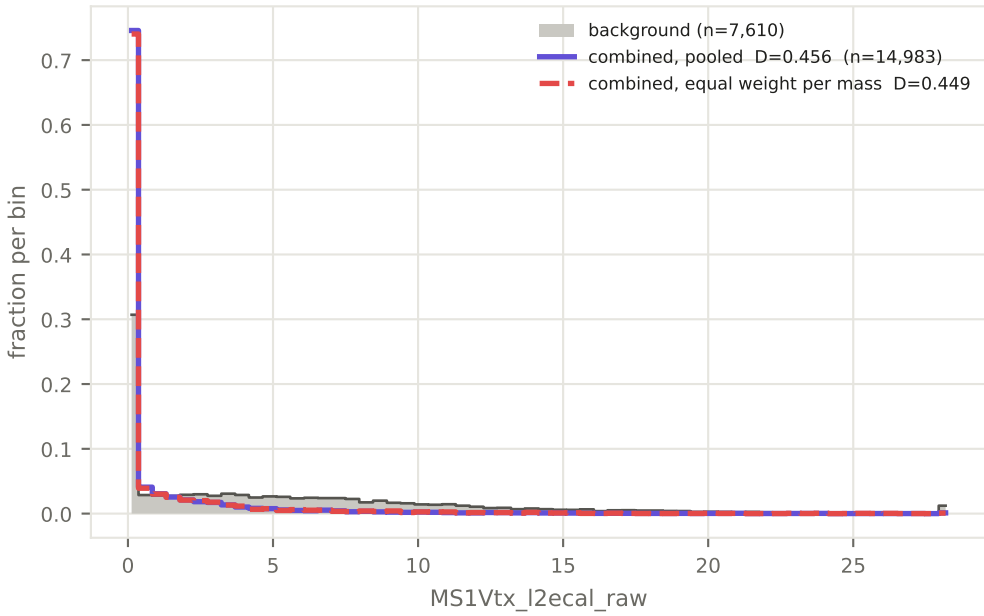
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



edge bins include under/overflow · NaN and -999 removed

MS1Vtx_l2ecal_raw

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 47 / 169

D(below vs above) 0.117 n below = 28,449 n above = 7,610 large D = background shape changes across the cut

